

Biomonitoring of urinary metabolites of 2-amino-1-methyl-6-phenylimidazo[4,5-b]pyridine (PhIP) following human consumption of cooked chicken.

Human risk assessment of exposure to 2-amino-1-methyl-6-phenylimidazo[4,5-b]pyridine (PhIP) through the diet may be improved by conducting biomonitoring studies comparing metabolism in humans and rodents. Eleven volunteers ingested a meal of cooked chicken containing 4'-OH-PhIP and PhIP in amounts of 0.6 and 0.8 microg/kg, respectively and urine was collected for the next 16h. The large number of PhIP metabolites was by treatment of the urine samples with hydrazine hydrate and hydrolytic enzymes reduced to three substances, 4'-OH-PhIP, PhIP and 5-OH-PhIP of which the first is a biomarker for detoxification and the last a biomarker for activation. The eleven volunteers eliminated large amounts of 4'-OH-PhIP in the urine. The majority of which could be accounted for by the presence of 4'-OH-PhIP in the fried chicken, showing that PhIP only to a small extent (11%) was metabolised to 4'-OH-PhIP. A larger fraction of the PhIP exposure, 38%, was recovered as PhIP and the largest fraction (51%) was recovered as 5-OH-PhIP suggesting that PhIP in humans to a large extent is metabolised to reactive substances. In rats, less than 1% of the dose of PhIP was eliminated as 5-OH-PhIP, suggesting that human cancer risk from exposure to PhIP is considerable higher than risk estimations based on extrapolation from rodent bioassays.